# HT-Advisor technician heat-treatment sheet

Fatigue objective - LPBF Inconel 718 - non-HIP route

## Input conditions

| Field | Value |
|---|---|
| Primary objective | fatigue |
| Alloy and process | LPBF Inconel 718 |
| Initial material state | EOS-like LPBF, as-built |
| Build orientation | vertical |
| Surface condition | machined |
| Representative section size | thin coupon |
| HIP benchmark included | No |
| Maximum furnace temperature | 1100 C |
| Maximum practical cycle time | 20 h |
| Fatigue validation target | R = 0.1; Nf = 1,000,000 cycles |

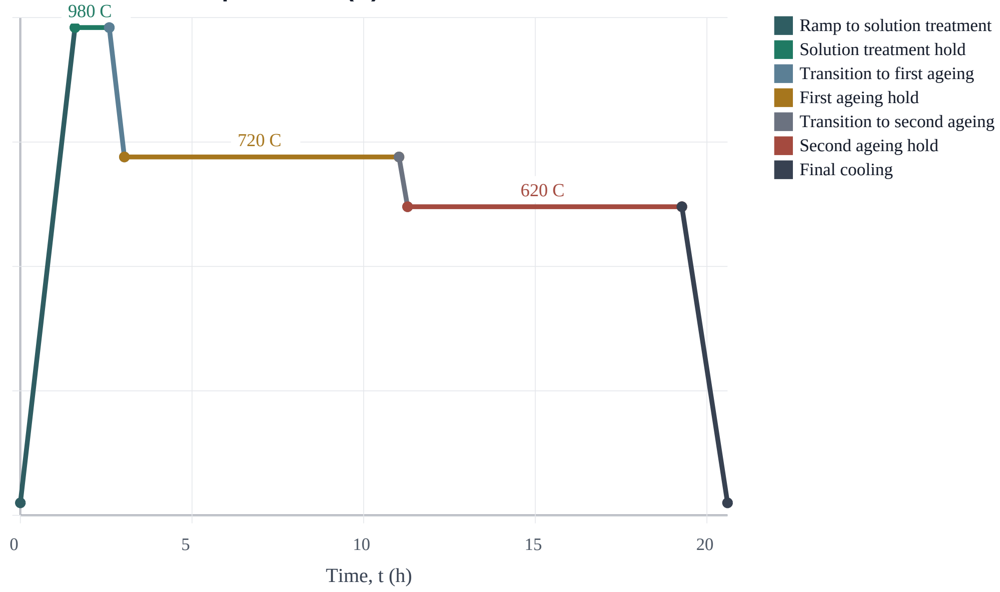
## Recommended route

| Field | Value |
|---|---|
| Route | ST_DA |
| Approved recipe | 980 C for 1 h; 720 C for 8 h; 620 C for 8 h |
| Peak temperature | 980 C |
| Total specified hold time | 17.0 h |
| Recommendation index | 0.77 |
| Local feasibility | feasible under selected constraints |

## Nominal thermal programme

| Step | Set point | Hold time | Action |
|---|---|---|---|
| 1 | 980 C | 1.0 h | Hold at temperature |
| 2 | 720 C | 8.0 h | Hold at temperature |
| 3 | 620 C | 8.0 h | Hold at temperature |

## Thermal step profile: temperature T (C)




The thermal profile is nominal. Record actual ramp rate, soak start and end times, cooling condition, specimen placement, and thermocouple or witness-coupon position.

# Fatigue-test instruction sheet

Use this page with the local fatigue-machine standard operating procedure. It defines the validation stresses for the recommended heat-treatment route; it does not replace machine-specific safety checks, fixture approval, or laboratory sign-off.

## Controller stress targets

| sigma_a (MPa) | R | sigma_max (MPa) | sigma_min (MPa) | sigma_mean (MPa) | Target / interpretation |
|---|---|---|---|---|---|
| 300 | 0.1 | 667 | 67 | 367 | 1000000 cycles; Primary runout candidate |
| 350 | 0.1 | 778 | 78 | 428 | 1000000 cycles; Boundary validation level |
| 400 | 0.1 | 889 | 89 | 489 | 1000000 cycles; Finite-life screen; not a runout expectation |
| 450 | 0.1 | 1000 | 100 | 550 | 1000000 cycles; Finite-life screen; not a runout expectation |

Stress ratio definition: R = sigma_min / sigma_max. Stress amplitude: sigma_a = (sigma_max - sigma_min) / 2. Mean stress: sigma_mean = (sigma_max + sigma_min) / 2.

Force conversion for the machine controller: Fmax = sigma_max x A0 and Fmin = sigma_min x A0, where A0 is the measured net cross-sectional area in mm2. The resulting force is in N when stress is in MPa.

## Recommended test sequence and review points

| Order | sigma_a | sigma_max | sigma_min | sigma_mean | Instruction |
|---|---|---|---|---|---|
| 1 | 300 MPa | 667 MPa | 67 MPa | 367 MPa | Primary runout candidate; perform first to check whether the selected recipe and specimen condition are viable. |
| 2 | 350 MPa | 778 MPa | 78 MPa | 428 MPa | Second validation level; continue only if 300 MPa reaches runout or gives scientifically useful finite life. |
| 3 | 400 MPa | 889 MPa | 89 MPa | 489 MPa | High-stress finite-life screen; use to populate the S-N curve if specimens and machine time allow. |
| 4 | 450 MPa | 1000 MPa | 100 MPa | 550 MPa | Highest-stress screen; run last and expect possible early fracture. |

- Start with 300 MPa. If the specimen fractures far below the target runout, pause and review surface finish, porosity evidence, alignment, and heat-treatment records before consuming higher-stress specimens.

- Run 400 MPa and 450 MPa only when finite-life data are required for fitting the local S-N curve or when sufficient spare specimens are available.

- Use the same specimen geometry, surface condition, build orientation, waveform, and frequency across the validation set unless the research lead approves a controlled comparison.

## Stop criteria

- Stop at specimen fracture.

- Stop at 1,000,000 cycles and record the result as runout if fracture has not occurred.

- Stop and flag the test if grip slip, load-control instability, fixture contact, abnormal displacement drift, machine alarm, or visible specimen heating occurs.

- Do not restart a stopped fatigue test on the same specimen unless the research lead records the reason and approves the restart.

# Three-specimen fatigue test reference

Use this sheet when only three fatigue specimens are available for the ST_DA validation route. The aim is to obtain the highest-value evidence: one likely runout point, one boundary point, and one finite-life point if the first two results permit it.

## Initial test plan

| Specimen | sigma_a | R | sigma_max | sigma_min | Purpose |
|---|---|---|---|---|---|
| 1 | 300 MPa | 0.1 | 667 MPa | 67 MPa | Primary runout candidate at 1,000,000 cycles. |
| 2 | 350 MPa | 0.1 | 778 MPa | 78 MPa | Boundary validation level after the 300 MPa result. |
| 3 | 400 MPa | 0.1 | 889 MPa | 89 MPa | Finite-life point if 300 and 350 MPa both reach runout. |

## Adaptive selection for the final specimen

| Observed result | Next action |
|---|---|
| 300 MPa reaches runout | Test 350 MPa next. |
| 300 MPa fractures well below target | Pause testing. Review surface finish, alignment, porosity evidence, and heat-treatment record before using the remaining specimens. |
| 350 MPa reaches runout | Use the final specimen at 400 MPa. |
| 350 MPa fractures before runout | Use the final specimen at 325 MPa to bracket the transition near 1,000,000 cycles. |
| 300 and 350 MPa both reach runout | Use 400 MPa to obtain a finite-life point for the local S-N trend. |

## Technician notes

- Run the specimens in sequence: 300 MPa first, then 350 MPa if the first result is acceptable.

- Do not continue to a higher stress level after an unexpectedly early fracture until the research lead has reviewed the specimen and setup records.

- Keep heat treatment, build orientation, surface condition, geometry, waveform, frequency, and room-temperature test condition unchanged across the three specimens.

- Record runout and fractured specimens separately. A runout at 1,000,000 cycles is not a measured failure life.

# Test records and validation controls

## Required heat-treatment records

- Furnace ID: to be completed.

- Furnace programme ID: to be completed.

- Furnace atmosphere, vacuum, or shielding gas: to be completed.

- Thermocouple or witness coupon location: to be completed.

- Final cooling method: controlled furnace cooling unless locally approved otherwise.

- Operator sign-off and process-owner approval: to be completed.

## Required fatigue-test records

- Specimen ID, heat-treatment batch ID, build orientation, surface condition, and measured roughness if available.

- Measured gauge dimensions and net cross-sectional area A0 used for load conversion.

- Machine ID, load cell range, fixture/grip type, alignment check, waveform, frequency, room temperature, and operator name.

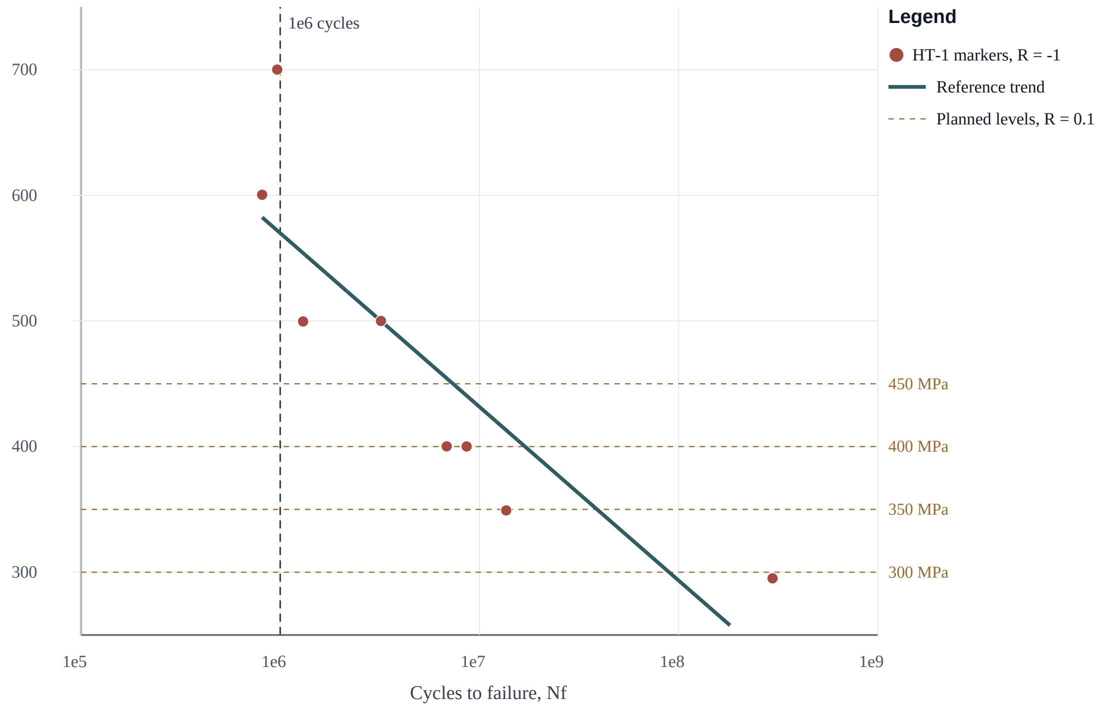
- Programmed sigma_a, sigma_max, sigma_min, sigma_mean, R, Fmax, Fmin, and target runout.

- Actual cycles to fracture or runout, failure location, fracture-surface photographs, and notes on grip slip or abnormal heating.

# Illustrative S-N curve context

### Reference markers and planned validation levels

Reference HT-1 data are from Wang et al. (2022), R = -1; proposed test levels are R = 0.1.




This plot is included to show the expected form of the fatigue dataset and the relative placement of the proposed stress levels. It should not be used to claim fatigue life for the present R = 0.1 tests because the reference markers were obtained at R = -1.

After testing, replace the planned-level guide lines with the measured local specimen points and report runout specimens separately from fractured specimens.

# Literature context and interpretation limits

## Reference evidence

Closest reviewed fatigue source: Wang et al. (2022), Journal of Alloys and Compounds 913:165171, DOI 10.1016/j.jallcom.2022.165171, Fig. 14. The paper reports selective laser melted Inconel 718 tested at room temperature using R = -1. Its HT-1 route is ST+DA-like but not identical to the present ST_DA recommendation.

| Stress amplitude | Closest ST+DA-like literature context | Use in this work |
|---|---|---|
| 300 MPa | Wang et al. (2022) report 8.41e7 cycles for HT-1 at R = -1. | Runout candidate at R = 0.1; test first. |
| 350 MPa | Digitised HT-1 marker: 1.36e7 cycles at R = -1. | Boundary high-cycle validation level. |
| 400 MPa | Digitised HT-1 markers: 6.85e6 to 8.63e6 cycles at R = -1. | High-stress finite-life screen; do not assume runout. |
| 450 MPa | Interpolated HT-1 context: approximately 3.0e6 cycles at R = -1. | Optional high-stress finite-life screen after lower levels. |

## Mean-stress caution

The present validation schedule uses R = 0.1, which introduces tensile mean stress. The Wang et al. values above are therefore reference context only, not fatigue-life predictions for these specimens.

| sigma_a | sigma_mean at R = 0.1 | Approx. Goodman-equivalent R = -1 amplitude | Interpretation |
|---|---|---|---|
| 300 MPa | 367 MPa | 418 MPa, assuming UTS = 1300 MPa | possible runout candidate |
| 350 MPa | 428 MPa | 522 MPa, assuming UTS = 1300 MPa | increasing finite-life risk |
| 400 MPa | 489 MPa | 641 MPa, assuming UTS = 1300 MPa | likely finite-life/rapid-failure regime |
| 450 MPa | 550 MPa | 780 MPa, assuming UTS = 1300 MPa | likely finite-life/rapid-failure regime |

- Treat the Goodman-equivalent values as a screening calculation only. Actual fatigue life can vary strongly with internal porosity, lack-of-fusion defects, surface condition, residual stress, and build orientation.

- Because the route is non-HIP, scatter between nominally identical specimens should be expected. Report failed specimens and runout specimens separately.

## Must-have experimental validation

- required: as-built baseline - Quantifies the starting LPBF Inconel 718 response before any thermal treatment.

- required: AMS-style standard baseline - Compares the framework-recommended route against a conventional solution plus double-ageing reference route.

- required: framework-recommended route: ST_DA - Tests the route selected by the decision-support workflow under the local furnace constraints.

- required: hardness and tensile testing - Validates the static property indicators used by the current metallurgy-informed model.

- required: SEM/EDS microstructural assessment - Checks Laves/Nb-rich segregation, precipitate response, and evidence that the selected heat treatment produced the intended metallurgical change.

- strongly recommended: fatigue or staircase screening if machine time is available - Non-HIP LPBF fatigue is defect-controlled; static tensile indicators alone do not establish fatigue safety.

## S-N model boundary

A literature-only right-censored Basquin S-N module has been trained from reviewed marker points at R = -1 or with stress ratio not reported. Runout markers are retained as right-censored lower-bound observations. No R = 0.1 local fatigue-life predictor is trained yet. The validation schedule above is intended to generate local fatigue data for calibration; it should not be cited as a trained local fatigue-life prediction.